PolicyCut

Write your privacy policy in plain English — review, redact and audit your video, fully local on AMD Radeon.

AMD Radeon Hackathon 2026

Track 1 · Multimodal Content Creation Tools

The problem

Real footage is full of personal data: bystander faces, plates, ID cards, screens, spoken phone numbers and addresses. Manual redaction takes hours; cloud services require uploading exactly the footage you must protect; single-model demos cover one modality with no review and no audit.

One sentence in, evidence out

```
"Pixelate every face, license plate,
 identity document and screen.
 Beep phone numbers, email addresses
 and street addresses."
```

Upload video (≤60 s)

▼

Interpret policy → structured rules you confirm

▼

Automatic analysis (vision + audio, on Radeon)

▼

HUMAN REVIEW — confirm / fix / add on a dual-track timeline

▼

Render → redacted MP4 + redacted SRT + audit.json

All local, all on Radeon

Policy → JSON rules
Qwen3-4B-Instruct, strict schema, deterministic keyword fallback

Open-vocab detection
Grounding DINO tiny (Transformers port — no CUDA custom ops)

Small-face assist
YuNet (OpenCV Zoo, CPU assist)

Mask tracking
SAM 2.1 hiera-small, video propagation, SAM2_BUILD_CUDA=0

Speech-to-text
Whisper large-v3-turbo, word-level timestamps

Spoken-PII locate
regex + spoken-form normalization ("five five five … at example dot com")

Fail toward privacy

- Empty SAM mask → padded box, never a hole
- Mask dilate 5–12 px + feather + safety box
- No word timestamps → whole segment muted
- Audio spans padded −150/+250 ms, gaps merged
- Unknown policy terms surfaced, never guessed
- Export blocked until every event reviewed

A real editor, not a demo page

Review workspace:

- video + live canvas overlay of tracked boxes
- dual-track timeline (visual + audio lanes)
- confirm / ignore / retime every event
- manual fixed regions for anything missed
- frame stepping, 0.5× rate, event hopping

Every export ships evidence

```
redacted.mp4   H.264/AAC, metadata stripped
redacted.srt   [REDACTED: PHONE|EMAIL|ADDRESS]
audit.json     models, precision, timings, VRAM,
               every event + human edits,
               zero PII plaintext inside
SHA-256 of input and output, ZIP bundle
```

Measured on Radeon PRO W7900

GPU: AMD Radeon PRO W7900 (48 GB)
ROCm/HIP: None
PyTorch: 2.11.0
Precision: FP16 (per-stage FP32 fallback)

Grounding DINO keyframes/s measured — see artifacts/
SAM 2.1 frames/s benchmarks/*.json and
Whisper real-time factor docs/benchmark.md for the
End-to-end real-time factor full measured tables
Peak VRAM per stage (real numbers, no estimates)

Honest Radeon engineering

- --require-rocm: core stages refuse silent
- CPU fallback — a Radeon run is provable
- Sequential model load/unload per stage
- torch.cuda.synchronize() around all timings
- Peak VRAM recorded per stage in audit.json
- ROCm-clean models: no CUDA custom kernels
- Zero external API calls during processing

PolicyCut is a human-in-the-loop privacy editor with an auditable, fully local workflow — not a guarantee of legal compliance.